

ARTICLE



Incidence and survival of uveal melanoma in Northern Ireland: how incomplete data can skew results in rare cancers

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BACKGROUND: The majority of Northern Irish uveal melanoma (UM) patients are diagnosed in Sheffield. This study aims to present incidence and survival outcomes for UM patients from Northern Ireland (NI).

METHODS: Collaborative retrospective study between Sheffield and Northern Ireland Cancer Registry (NICR). For UM cases not on both databases, outcomes and survival rates (via Kaplan-Meier analysis) were compared. Anonymised NICR data were used to calculate whole-population incidence of UM for NI.

RESULTS: In total, 161 patients from NI were diagnosed in Sheffield, 90 of which were not registered with NICR at the start of this study. Data-omissions were not consistent across patient groups, leading to significant differences between those patients registered and those not. Registered patients had an all-cause 5-year survival rate of only 68.9% compared to 92.5% of those not registered ($p < 0.01$) and were $>17\times$ more likely to have systemic metastases than those not registered ($p < 0.001$). Following rectification of data-omissions, the European age-standardised incidence rate of UM for NI was 8.6 per million.

CONCLUSIONS: This study illustrates the impact of incomplete population-wide data, serving as a real-world lesson in case-identification bias. Rare cancers are at higher risk of omission due to systemic failures as the small numbers involved are not detected by system-wide validation procedures. Following this study, data-transfer agreements between England and NI were actioned, preventing future data-omissions. We present survival and incidence data for UM in NI for the first time, showing the incidence is amongst the highest in Europe, with good survival rates.

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INTRODUCTION

With an incidence between 1.3 and 8.6 per million in European populations, appearing to increase with higher geographical latitudes, uveal melanoma (UM) is a rare cancer. [1] Within the United Kingdom (UK), rates were lower in England and Wales (3.2–6.4) than in more northern Scotland (8.4 per million). [1] The incidence in neighbouring Republic of Ireland (ROI) is notably higher than would be expected, at 9.5 per million. [2] Rates for Northern Ireland (NI), the fourth and final component nation of the UK, have never been reported.

Despite advancements improving detection and treatment of UM, the reported 5-year relative survival rate has remained stable [3] with a mortality of around 30% at 5 years. [3, 4] Such figures are calculated from population-wide cancer registries, the accuracy of which are heavily reliant on the completeness of these databases. [5] In rare cancers such as UM, a relatively small absolute number of omissions could have significant effects on analyses such as survival rates, especially if case-losses are inconsistent across patient groups.

The Northern Ireland Cancer Registry (NICR) covers the population of NI, which is around 1.9 million, [<https://www.nisra.gov.uk/sites/nisra.gov.uk/files/publications/MYE20-Bulletin.pdf>] and aims to

hold data on all patients who have been diagnosed with cancer since 1993. NICR uses multiple sources of information: histopathological reports, hospital databases and statutory death certifications. Case ascertainment is achieved via a combination of automated receipts through data transfer agreements and active case verification by registry staff. A recent validity and accuracy study of NICR indicated a ‘very high’ level of accuracy and completeness by comparing NICR data with records held independently by general practitioners. [6]

UM is treated outside the local healthcare pathways in NI: there is no specialist ocular oncology provision locally, so patients are referred to one of three national centres in England. This system has been in place for decades. Sheffield Ocular Oncology Service (SOOS) is one such centre, seeing up to two NI patients per month in recent years. Data transfer agreements exist between regional UK cancer registries related to persons treated outside their area of residence, designed to capture all cancer cases regardless of where the patient was diagnosed in the UK.

No studies have published data describing the epidemiology of NI patients with UM. Given that NI and ROI share ethnic similarities and the same landmass, one might expect similar or higher rates based on geographical latitude.

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We established a bilateral data-sharing agreement in 2020 between SOOS and NICK regarding UM patients. The purpose of this study was to investigate the incidence of UM patients in NI, and to report clinical outcomes including recurrence, metastatic and survival rates. On comparing SOOS data with NICK, we found that over half of those patients treated by SOOS were initially missing from the NICK. Further study went on to examine the effect of this loss of data. Finally, once all cases of UM in NI were identified, the population-incidence was investigated.

METHODS

Patients with UM from NI diagnosed in Sheffield between January 1998 and December 2018 were identified from the Sheffield Ocular Oncology Database (Infoflex Digital health software, Civica, London), which has been manually maintained by dedicated information coordinators. Pertinent patient data were collected by retrospective case note review. Data collected included patient demographics at diagnosis, site and laterality of tumour, basis of diagnosis, tumour dimensions, anatomical category of primary tumour, [7] date and type of treatment, tumour recurrence or metastases, and any further treatment.

Sheffield ocular oncology service

All cases of suspected UM from NI are assessed by their local ophthalmologist and referred to Sheffield based on referrer preference. UM was diagnosed based on clinical and imaging features; diagnostic biopsy is infrequently performed in line with the majority of ocular oncologists in the UK. [8] Biopsy is reserved for atypical cases or for prognostication. Patients clinically diagnosed with UM at SOOS were discussed at the multidisciplinary team (MDT) meeting, the minutes of which are the primary data-source for the Sheffield database. MDT discussion is a statutory requirement, ensuring robust case capture. If patients were unable to travel (for example due to poor general health), clinical images taken locally in NI could be shared and reviewed electronically at MDT, resulting in capture by the Sheffield database.

Once diagnosed with UM, patients were offered treatment based on tumour suitability, patient preference and fitness for general anaesthetic. Modalities included laser-based local treatment (photodynamic therapy, transpupillary thermotherapy), radiotherapy (ruthenium plaque brachytherapy, proton beam therapy, stereotactic radiosurgery) and surgery (local resection or enucleation). Where appropriate, treatment by enucleation was performed locally by ophthalmologists in NI, all other treatments were administered in Sheffield. Those who declined treatment were captured by the Sheffield database as inclusion was based on diagnosis not treatment. Following treatment, patients were referred back to NI for systemic metastatic surveillance after initial screening by means of abdominal ultrasound in Sheffield.

Northern Ireland cancer registry

Data from UM patients with an address in NI were shared with NICK, and through the reciprocal agreement, data including the date and cause of death, and information regarding systemic metastases were collected from NICK in October 2020.

Calculating incidence

Anonymised population-wide data from NICK, following comprehensive data-transfers between regional cancer registries in the UK identifying all UM cases from NI, were used to calculate the European age-standardised incidence rate covering the 21-year inclusion period (January 1998–December 2018). Throughout this report the 2013 European Standard Population was used. UM cases were identified by searching the NICK by ICD-O-3 [9] codes, and the associated morphology codes [10] 8720–8774/3 indicating melanoma. Codes C69.3 and C69.4 refer to neoplasms of the choroid and ciliary body, or iris respectively. These cases are all UMs when associated with the appropriate morphology code. C69.8 (neoplasm of overlapping sites of eye and adnexa) and C69.9 (neoplasm of unspecified site of eye) are also included in line with other population-wide studies of UM, [1, 3, 4] though rarely cases with these codes are not uveal in origin (for example, conjunctival).

Statistical analysis

Data analysis was carried out using IBM SPSS Statistics version 27.0 (IBM Corp., Armonk, NY, USA). Data approximating to a normal distribution were

described using means and standard deviations (SD), whereas non-normal data were described using medians and inter-quartile ranges (IQR). Comparisons were made using independent *t* tests, Mann-Whitney-*U* tests or Chi-squared tests as appropriate. Survival analysis was carried out using the Kaplan-Meier method with comparisons between groups made with the log-rank test. Cases were censored at the end of their follow-up for any reason other than death for all-cause mortality analyses, and for end of follow-up or non-melanoma death for the melanoma-associated mortality analysis. The influence of competing causes of deaths was therefore ignored for the melanoma-associated death analysis. The start date for analyses was taken as the date of treatment of UM, and where no treatment was given, the date of diagnosis was used. The date of treatment was chosen due to discrepancies in the way NICK records date of diagnosis: NICK prioritises histological diagnosis over clinical, so in cases treated by enucleation, the date of diagnosis documented on NICK is the date of histological confirmation, which was several weeks/months after the clinical diagnosis was made.

RESULTS

All Sheffield-diagnosed Northern Irish patients

In total, 161 patients from NI were diagnosed with UM in Sheffield, from January 1998 to December 2018. The demographics of the cohort can be found in Table 1. Electronic and paper records were available for all patients. All patients were diagnosed based on clinical opinion, as no patients requested prognostic biopsy. No patients had evidence of metastatic disease at diagnosis.

All but four patients elected to undergo treatment of their primary tumour, the details of which are found in Table 1. Of those treated, 78% patients were given eye-sparing treatments. Tumour recurrence occurred in ten patients (6.2%); all of whom had initially received eye-sparing therapy. All recurrences underwent secondary treatment: two of which were enucleated. Three patients experienced a second recurrence requiring enucleation to achieve local tumour control. Overall, 47 patients (29.2%) had eventual histological confirmation of diagnosis from surgical treatment (enucleation, *n* = 46, or local resection, *n* = 1). The majority of patients did not have any histology from the primary tumour in keeping with trends globally. [10]

The median follow-up duration of the Sheffield cohort was 84.0 months (IQR 57.5–132.5). During which, 27 (14.8%) patients developed systemic metastases from UM, detected a median of 23.0 months (IQR 13.0–40.0) after initial treatment.

During the follow-up period, 43 (26.7%) patients died; of these 16 (37.2%) had UM listed as the primary cause of death and 25 (58.1%) had melanoma listed as a contributory factor on the death certificate. Although not explicit, it is presumed that for UM to have contributed to a death (and therefore be listed on the certification), that there must have been UM metastases present. 114 (70.8%) were alive as of October 2020, the mortality status was unknown for 4 patients (2.2%). The all-cause 5-year survival across all NI patients treated in Sheffield for UM was 82.6%. Figure 1 shows the Kaplan-Meier survival plot and analysis for all-cause mortality (including death from all causes). The Kaplan-Meier analysis of all-cause mortality by TNM stage is included in Supplemental Fig. 1.

Comparison of Sheffield data with NICK

Of the 161 patients identified by Sheffield, only 71 (44.1%) were already registered as having UM on NICK, hereafter referred to as the 'NICK-known' group (Fig. 2A). NICK-known patients were compared to those patients who, at the beginning of this study, were not registered with NICK, but were captured by 'Sheffield-only.' No statistically significant difference between the two groups was found comparing patient demographics, tumour site and follow-up duration (Table 2).

However, there were significant differences in tumour characteristics and clinical outcomes between the two groups. When compared to the Sheffield-only group, NICK-known patients had

Table 1. Patient demographics and lesion characteristics for whole Sheffield cohort ($n = 161$).

	Number of Patients (%)
Mean age at diagnosis in years [SD]	58.2 [14.5]
Sex	
Male	75 (46.6)
Female	86 (53.4)
Laterality of uveal melanoma	
Right	83 (51.6)
Left	78 (48.4)
Site of uveal melanoma	
Iris	4 (2.5)
Iridociliary	3 (1.9)
Ciliary Body	8 (5.0)
Ciliochoroidal	14 (8.7)
Choroidal	132 (82.0)
Tumour category [7] at diagnosis	
cT1	65 (40.4)
cT2	55 (34.2)
cT3	32 (19.9)
cT4	9 (5.6)
Median tumour thickness in mm [IQR]	3.9 [2.5–6.5]
Mean maximum basal tumour diameter in mm [SD]	10.7 [3.1]
Median follow-up in months [IQR]	84.0 [57.5–132.5]
Primary treatment of uveal melanoma	
None	4 (2.5)
PDT	8 (5.0)
TTT	3 (1.9)
Ruthenium plaque	57 (35.4)
Stereotactic	31 (19.3)
Proton Beam	18 (11.2)
Local resection	1 (0.6)
Enucleation	39 (24.2)

PDT photodynamic therapy, TTT transpupillary thermotherapy, SD standard deviation, IQR interquartile range (Mean and SD for normally distributed data and median and IQR for non-normal data).

significantly larger tumours, of median thickness of 4.6 mm cf. 3.5 mm ($p = 0.01$) and mean maximum diameter of 11.9 mm cf. 9.5 mm ($p < 0.001$). Correspondingly, NICKR-known tumours were of higher AJCC [7] tumour categories: 31.9% of NICKR-known patients were category cT1 cf. 47.2% of Sheffield-only ($p = 0.002$), 27.0% of NICKR-known were cT2 cf. 39.8% of Sheffield-only ($p = 0.01$), 28.4% were cT3 cf. 13.0% of Sheffield-only ($p = 0.03$) and 13.5% of NICKR-known tumours were cT4 cf. with 0% of Sheffield-only patients. NICKR-known patients were far more likely to have been treated by enucleation (45.1%) than Sheffield-only patients (8.9%), $p < 0.001$. NICKR-known patients also experienced a higher rate of metastasis in 35.2% compared with only 2.2% of Sheffield-only patients ($p < 0.001$).

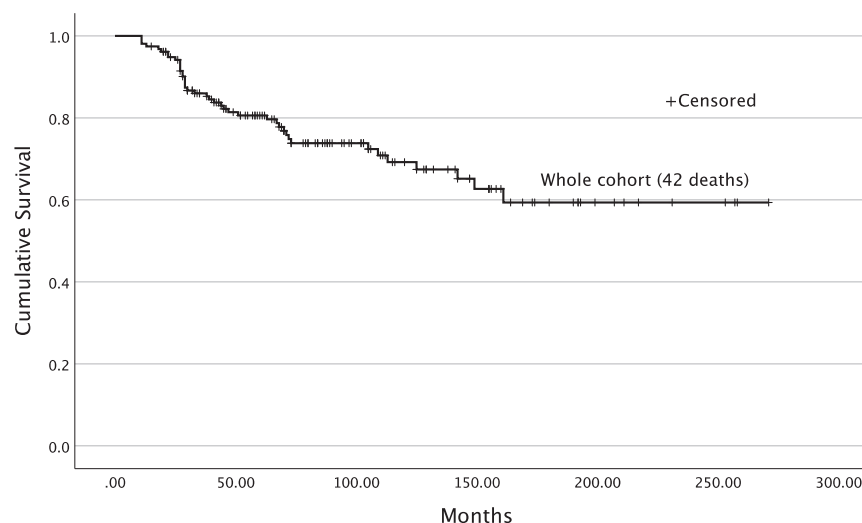
Figure 3 shows the Kaplan-Meier overall survival plot (death from all-causes) comparing the Sheffield-only cohort with the NICKR-known cohort. The mean estimated survival time was 19.7 years [95% confidence interval (CI) 18.0–21.5] for the Sheffield-only cohort and 11.0 years [95% CI 8.6–13.3] for the NICKR-known cohort. The all-cause 5-year survival rate was markedly different between the NICKR-known and Sheffield-only patients, at 68.6% and 92.3%, respectively.

When considering melanoma-related deaths only, i.e., those patients where UM was documented as a cause during death-certification, the mean estimated survival time for NICKR-known patients was 13.8 years [95% CI 11.4–16.3] and in the Sheffield-only group was 22.1 years [95% CI 21.3–2.8]. A log-rank (Mantel-Cox) comparing the groups showed this to be significant ($\chi^2 27.3$, $p < 0.01$).

Incidence of uveal melanoma in Northern Ireland

Following this study, NICKR investigated how some Sheffield-diagnosed patients were initially not registered. Pre-existing data-transfer agreements between regional cancer registries across the UK were found to have not been actioned. Once rectified, all 161 cases diagnosed by Sheffield plus a further 103 UM cases diagnosed elsewhere were identified. 36 patients with non-ocular tumour diagnoses (21 cases of in situ disease or tumours of “uncertain behaviour”, seven cases of prostate carcinoma and eight miscellaneous cancer diagnoses) were also identified who were not already registered with NICKR, improving the fidelity of the NICKR for other specialties.

In total, 264 cases of UM were diagnosed in NI patients during the inclusion period (Fig. 2B), giving an age-standardised incidence rate of 8.6 per million (95% CI 7.5–9.6). The incidence

**Fig. 1** A Kaplan-Meier analysis of all-cause mortality for all 161 patients diagnosed in Sheffield.

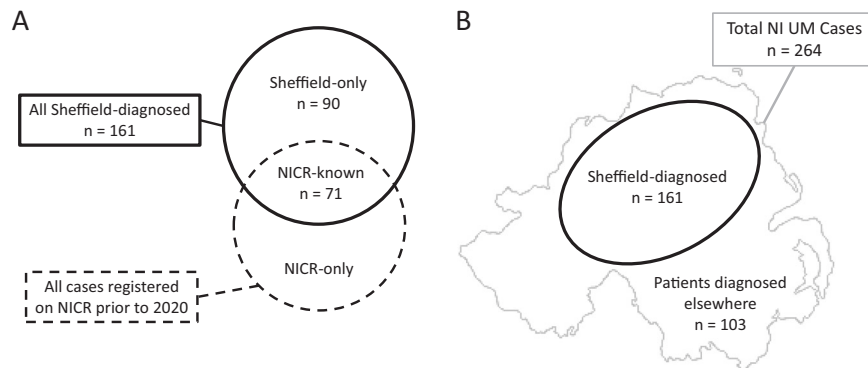


Fig. 2 Venn diagrams illustrating the groups described in this study, and the number of cases in each group. **A** at the initiation of the study, prior to the rectification of missing data. **B** after data-transfer between regional cancer registries and rectification of data omissions. NICTR Northern Ireland Cancer Registry, NI Northern Ireland, UM uveal melanoma.

was higher for men, at 9.9 per million (95% CI 8.2–11.6), than for women, at 7.6 per million (95% CI 6.3–9.0).

If incidence had been calculated from NICTR data before the resolution of data-omissions, at least 90 cases (or 34%) would not have been included which would have caused a significant underestimation of incidence.

DISCUSSION

National cancer registries, such as NICTR, aim to hold data for all cancers diagnosed in their population, and in so doing, facilitate research and aid planning of future cancer care. It is vital therefore, that databases are as accurate and complete as possible, with robust mechanisms to ensure case capture. Indeed, NICTR published on its overall high levels of accuracy, ascertained by comparing its data with records held independently by primary care. [6] The discovery of significant numbers of UM patients not registered on NICTR was therefore unexpected and concerning. Previous studies have also reported under-enumeration of ocular cancers by population-wide registries, in particular when comparing to ocular oncology records [11, 12]. However, the previously reported levels of under ascertainment were much lower than that found in our study. It is important to understand the causes of the missing UM patients in this study, to draw out lessons for population-based cancer registration and improve ongoing data fidelity.

As mentioned previously, there are three main sources of data for NICTR, two of which were found to be of limited efficacy for ocular cancers. Considering electronic hospital records, UM patients were treated via an atypical healthcare pathway outside NI, meaning that NICTR could not access healthcare-episode data at the time of diagnosis, preventing automated registration. In contrast, metastatic screening was undertaken locally, resulting in those with metastatic disease being preferentially registered. This culminated in the metastatic rate of NICTR-known cases being double that of the “whole Sheffield cohort”, and more than 17x higher than those patients who were previously known to “Sheffield-only”. This skewed survival rates, as these patients were more likely to die overall, and from UM specifically.

NI patients on atypical healthcare pathways had been anticipated as a potential problem for cancer registration in the past, albeit for a very small number of patients. Data-transfer agreements had been established between regional and national cancer registries to capture these patients. Unfortunately, these had not been enacted for some time prior to this study, leading to higher-than-expected levels of under ascertainment by NICTR. A major factor in the failure of automated data-transfer was found to

be changes in data protection legislation in the UK. When the Data Protection Acts of 1998 and 2018 came into force, the automated data-transfers were stopped pending review and reapplication. This was overlooked until the present study identified missed NI cancer registrations on NICTR, and was remedied by reapplication and compliance with current data protection laws.

NICTR reliance on histopathology data, in the absence of hospital records, was also a source of under enumeration and bias. UM was predominantly diagnosed based on clinical opinion, in keeping with trends globally, [8] meaning that histology was unavailable for the majority of patients. Almost all patients treated surgically, and therefore registered via histology, were treated by enucleation. As enucleation is reserved for larger and higher stage lesions, [7] these patients have the highest risk for metastasis and death. [3, 13] The result was that the deadliest tumours were most likely to be registered, which was reflected in the significantly worse survival rate for NICTR-known patients.

A final and significant contributor to under ascertainment by NICTR, and in previous studies, [11, 12] is a consequence of the rarity of UM. In 2016 alone, there were 9446 new cancer diagnoses across NI. [https://www.macmillan.org.uk/_images/cancer-statistics-factsheet_tcm9-260514.pdf] With an incidence of 8.6 per million, UM would have represented less than 0.2% of all cancers diagnosed in the population that year. Even if all UM cases were missed from the registry, validity and completeness studies performed registry-wide [6] would not detect the omission relative to the total number of cancer registrations. Rare cancers are therefore at a higher risk of being missed by systematic failures as their absolute numbers are so small.

To improve retrospective and future cancer registrations, NICTR renewed the pre-existing data-transfer agreements with regional cancer registries, who were able to access hospital records and histopathology records for patients diagnosed in England. This single action resulted in the registration of 36 NI patients with other tumour diagnoses, as well as to the identification of all previously unregistered UM cases in NI, including the 90 “Sheffield-only” patients. The vulnerability of NICTR to data-losses from patients treated elsewhere has been highlighted by this experience, and systems are now in place to prevent such issues in future—especially when legislation changes occur affecting data-sharing. The authors are as confident as possible that all cases have now been identified and included in the incidence analysis. The regional cancer registries in England receive their data from multiple sources (similar to the NICTR) to corroborate and ensure as complete a dataset as possible. It is therefore extremely unlikely that any UM patients treated in England were missed from the English cancer registries. Now that all UM patients from NI registered on the English registries have been shared with NICTR,

Table 2. Patient demographics and lesion characteristics in NICR-known patients ($n = 71$) vs. Sheffield-only patients ($n = 90$).

	NICR-known [<i>n</i> = 71]	Sheffield-only [<i>n</i> = 90]	<i>p</i> value*
Mean age in years (SD)	59.9 (15.0)	57.9 (14.2)	0.39
Male sex, <i>n</i> (%)	30 (40.2)	45 (50.0)	0.33
Laterality right, <i>n</i> (%)	36 (50.7)	47 (52.2)	0.85
Site of uveal melanoma, <i>n</i> (%)			
Iris	1 (1.4)	3 (3.3)	NA
Iridociliary	2 (2.8)	1 (1.1)	
Ciliary Body	2 (2.8)	6 (6.7)	
Ciliochoroidal	11 (15.5)	3 (3.3)	
Choroidal	55 (77.5)	77 (85.6)	
Tumour category [7] at diagnosis, <i>n</i> (%)			
cT1	23 (31.9)	42 (47.2)	0.002
cT2	19 (27.0)	36 (39.8)	0.01
cT3	20 (28.4)	12 (13.0)	0.032
cT4	9 (12.7)	0	NA
TNM stage, <i>n</i> (%)			
I	22 (31.0)	45 (50.0)	<0.001
IIA	19 (26.8)	32 (35.6)	
IIB	6 (8.5)	9 (10.0)	
IIIA	15 (21.1)	4 (4.4)	
IIIB	7 (9.9)	0	
IIIC	2 (2.8)	0	
Median tumour thickness in millimetres [IQR]	4.6 [2.7–10.0]	3.5 [2.5–5.5]	0.01
Mean maximum basal tumour diameter in millimetres [SD]	11.9 [4.1]	9.5 [2.7]	<0.001
Development of metastases, <i>n</i> (%)	25 (35.2)	2 (2.2)	<0.001
Median follow-up in months [IQR]	84.0 [65.0–128.0]	87.5 [46.0–139.8]	0.61
Primary treatment of uveal melanoma			
None	1 (1.4)	3 (3.3)	NA
Globe-sparing treatment	39 (54.9)	79 (87.8)	<0.001
Enucleation	31 (43.7)	8 (8.9)	

SD standard deviation, NA not applicable, TNM stage—the Union for International Cancer Control TNM classification, NB: all patients were N0M0 in this study at diagnosis.

(**p* values refer to Mann–Whitney U test, independent *t* test or Chi squared test as appropriate, a *p* value of <0.05 is considered significant).

all patients should be accounted for. Further study with other ocular oncology centres would be beneficial to further corroborate the data, however.

With a European age-standardised incidence rate of UM of 8.6 per million, NI has one of the highest rates in Europe. Only Scotland, Norway, Denmark and Sweden have similarly high incidence rates, yet all are at higher geographical latitudes. [1] The reported rate in ROI [2] remains significantly higher than expected in comparison, though it is important to note that the ROI data were standardised to an older reference population (1976) [2].

Given the ethnic and geographical similarities between NI and ROI, [14] one might have expected more similar incidence rates between the two countries. The difference may be a function of the ROI [2] study reflecting only 5-years of data, in contrast to our study and others which span decades. [1, 2] Given the small absolute number of UM diagnoses, random variation may have resulted in higher incidence rates over a relatively short period of time. Further study is needed, and we look forward to updated data from the ROI group.

Overall, NI patients diagnosed in Sheffield with UM experienced excellent clinical outcomes: three quarters of patients were treated with eye-sparing treatments, with low recurrence rates and an all-cause 5-year survival rate of over 80%. Our results compare favourably with studies from the US [3, 15] and Europe. [4] Relative to the aforementioned, almost pan-continental studies, [3, 4, 15] the number of cases reported here is very small, but the results are nevertheless encouraging.

The Sheffield cohort does not include all UM cases registered in NI; 103 further cases are registered on NICR during the inclusion period (Fig. 2B). Cases not included on the Sheffield database fall into one of three groups: they were referred elsewhere (most likely), refused referral or were too unwell for referral. The authors believe that cases referred elsewhere are not likely to be from any particular patient or prognostic group, as it has been our observation that referral practices are based on referrer-preference and habit, not patient type. These patients are therefore unlikely to be a source of significant case-selection bias which would skew clinical outcomes.

In contrast, if the patient's health was so poor as to render referral or investigation futile, their absence from the Sheffield cohort could lead to overestimation of survival and may have contributed to the apparent difference between NICR-known and Sheffield-only groups. That being said, it is important to note that very few patients are likely to have been terminal: less than 2% of UM patients have systemic metastases at the time of diagnosis, which are typically asymptomatic and discovered only upon staging following UM diagnosis. [16–19] In this study population of 264 cases, fewer than 5 patients would have had detectable metastases at presentation, and fewer still are likely to have been significantly unwell. When patients were too frail to travel, remote consultations were available, especially in recent years. The authors believe the likelihood of a significant number of patients being excluded from the Sheffield cohort due to poor health or disseminated UM metastases is possible, but remote. That being said, no patients in the Sheffield-treated group ($n = 161$) had evidence of metastasis at presentation—statistically, three patients on average would have metastases at diagnosis. It is possible that patients with metastases were not referred to Sheffield, or perhaps random variation of small numbers accounts from the lack of patients with detectable metastases at presentation.

We report on the epidemiology of UM in NI for the first time, adding to the body of evidence from the island of Ireland, the UK and Europe as a whole, giving context to previous work. The process of this study identified significant numbers of patients who were not registered with the NICR, revealing the vulnerability of population-wide national cancer registries to case-identification bias. This finding has been reported previously in ocular cancers, [11, 12] but never at this magnitude and with such vivid illustration of the effects. Prior to this work, use of NICR-data alone would have wrongly implied that UM was rarer and deadlier in NI when compared to elsewhere in Europe. Our findings lend support to previous concerns regarding cancer registry completeness in the UK, [5] and to suggestions that differences in outcomes between countries may be more to do with omissions than actual differences in disease or prognosis. [19] Further study is needed, including but not limited to collaboration with other specialist centres to corroborate the NI data. Our experience and the

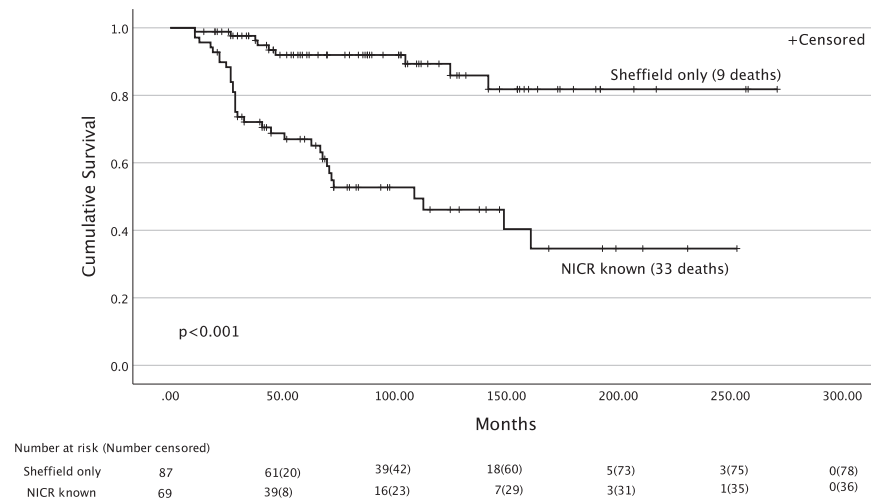


Fig. 3 A Kaplan-Meier analysis of all-cause mortality comparing the Sheffield-only cohort with NICTR-only patients. p value from log rank (Mantel-Cox) comparing the two groups (χ^2 27.2, $p < 0.002$).

analyses presented shine a spotlight on the potential impacts of incomplete data, especially in rare cancers, and serve as a reminder and a lesson for all.

SUMMARY

What was known before

- Uveal melanoma is a rare cancer, with an incidence of 1.3 to 8.6 per million in European populations (EUROCare), and even higher rates in the Republic of Ireland.
- Uveal melanoma in Northern Ireland has never been studied.
- Population-based studies are potentially vulnerable to data omissions and case selection bias.

What this study adds

- The incidence of uveal melanoma in Northern Ireland is 8.6 per million, with an 82.6% all cause 5-year survival rate. This data is published for the first time, improving the understanding of uveal melanoma in the UK and in Europe as a whole.
- Prior to this study, significant numbers of uveal melanoma cases were not captured on the national cancer registry
- Case-identification bias would have led to significant underestimation of incidence and survival rates, highlighting the importance of robust case-capture and data validation studies.

DATA AVAILABILITY

Northern Ireland Cancer Registry data are available and can be accessed by direct application. The data for the Sheffield cohort can be accessed by written application to the authors.

REFERENCES

1. Virgili G, Gatta G, Ciccolallo L, Capocaccia R, Biggeri A, Crocetti E, et al. Incidence of uveal melanoma in Europe. *Ophthalmology*. 2007;114:2309–15.
2. Baily C, O'Neill V, Dunne M, Cunningham M, Gullo G, Kennedy S, et al. Uveal melanoma in Ireland. *Ocul Oncol Pathol*. 2019;5:195–204. <https://doi.org/10.1159/000492391>.
3. Singh AD, Topham A. Survival rates with uveal melanoma in the United States: 1973–1997. *Ophthalmology*. 2003;110:962–5. [https://doi.org/10.1016/S0161-6420\(03\)00077-0](https://doi.org/10.1016/S0161-6420(03)00077-0).
4. Virgili G, Gatta G, Ciccolallo L, Capocaccia R, Biggeri A, Crocetti E, et al. Survival in patients with uveal melanoma in Europe. *Archives of ophthalmology*, 2008. 126. pp. 1413–8. ISSN 0003-9950 <https://doi.org/10.1001/archophth.126.10.1413>
5. Beral V, Peto R. UK cancer survival statistics. *BMJ*. 2010;341:c4112 <https://doi.org/10.1136/bmj.c4112>
6. Kearney TM, Donnelly C, Kelly JM, O'Callaghan EP, Fox CR, Gavin AT Validation of the completeness and accuracy of the Northern Ireland Cancer Registry. *Cancer Epidemiol*. 2015;39:401–4. <https://doi.org/10.1016/j.canep.2015.02.005>.
7. AJCC Cancer Staging Manual 8th ed. 2017, Corr. 3rd printing 2018 Edition
8. Nathan P, Cohen V, Coupland S, Curtis K, Damato B, Evans J, et al. Uveal melanoma UK national guidelines. *Eur J Cancer*. 2015;51:2404–12. <https://doi.org/10.1016/j.ejca.2015.07.013>. Epub 2015 Aug 13. PMID: 26278648
9. International classification of diseases for oncology (ICD-O), 3rd ed., 1st revision 3rd edition, 1st revision, <https://apps.who.int/iris/handle/10665/96612>
10. Stang A, Parkin DM, Ferlay J, Jöckel KH. International uveal melanoma incidence trends in view of a decreasing proportion of morphological verification. *Int J Cancer*. 2005;114:114–23. <https://doi.org/10.1002/ijc.20690>. PMID: 15523698
11. Vajdic CM, Krickler A, Giblin M, McKenzie J, Aitken J, Giles GG, et al. Incidence of ocular melanoma in Australia from 1990 to 1998. *Int J Cancer*. 2003;105:117–22. <https://doi.org/10.1002/ijc.11057>. PMID: 12672041
12. Iscovich J, Ackerman C, Andreev H, Pe'er J, Steinitz R. An epidemiological study of posterior uveal melanoma in Israel, 1961–1989. *Int J Cancer*. 1995;61:291–5. <https://doi.org/10.1002/ijc.2910610302>. 341 PMID: 7729936
13. Kaliki S, Shields CL. Uveal melanoma: relatively rare but deadly cancer. *Eye*. 2017;31:241–57. <https://doi.org/10.1038/eye.2016.275>
14. Gilbert E, O'Reilly S, Merrigan M, McGettigan D, Molloy AM, Brody LC, et al. The Irish DNA atlas: revealing fine-scale population structure and history within Ireland. *Sci Rep*. 2017;7:17199. <https://doi.org/10.1038/s41598-017-17124-4>. Erratum in: *Sci Rep*. 2018 May 3;8(1):7208.
15. Aronow ME, Topham AK, Singh AD. Uveal melanoma: 5-year update on incidence, treatment, and survival (SEER 1973–2013). *Ocul Oncol Pathol*. 2018;4:145–51. <https://doi.org/10.1159/000480640>
16. Garg G, Finger PT, Kivelä TT, Simpson ER, Gallie BL, Saakyan S, et al. Patients presenting with metastases: stage IV uveal melanoma, an international study. *Br J Ophthalmol Published Online First*: 15 January 2021. <https://doi.org/10.1136/bjophthalmol-2020-317949>
17. Steckler AM, Francis JH, Shoushtari AN, Abramson DH, Barker CA. Uveal melanoma metastatic at initial diagnosis: a case series. *Melanoma Res*. 2022;32:120–3. <https://doi.org/10.1097/CMR.0000000000000807>.
18. Rantala ES, Hernberg MM, Piperno-Neumann S, Grossniklaus HE, Kivelä TT Metastatic uveal melanoma: The final frontier. *Prog Retin Eye Res*. 2022;101041. <https://doi.org/10.1016/j.preteyeres.2022.101041>.
19. Robinson D, Sankila R, Hakulinen T, Möller H. Interpreting international comparisons of cancer survival: the effects of incomplete registration and the presence of death certificate only cases on survival estimates. *Eur J Cancer*. 2007;43:909–13. <https://doi.org/10.1016/j.ejca.2007.01.007>.

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AUTHOR CONTRIBUTIONS

HQ conceptualised the study, performed the literature search, study design, data collection, data analysis, data interpretation, produced the figures and wrote the original draft of the paper. MJM contributed to data analysis, interpretation, and verification of the underlying data, produced the figures, and reviewed and edited the paper. IGR, AG and PR contributed to conceptualisation and methodology and reviewed and edited the paper. DF and GS contributed to data collection, data analysis, verification of the underlying data and edited the paper. SS and DC contributed to the review and editing of the paper.

COMPETING INTERESTS

The authors declare no competing interests.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

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